

# Post-process User Guide

ATE log analysis → next-chip pattern recommendation · Verilumen Simulation Agent

This guide explains the **Post-process** window. Use it **after** ATE testing: take the fail log from **chip N** (one log = one die/chip), analyze which patterns failed, and get an **ordered list** of patterns to run on **chip N+1**.

## How the stages fit together

**Pre-process** (before ATE): STIL → keep/skip prediction. **ATE**: run patterns, produce a fail/pass log. **Post-process** (after ATE): log + last pre-process lists → ordered patterns for the next chip. Rule of thumb: **1 ATE log file = 1 die = 1 chip**.

## Recommended steps

- Step 1.** Finish **Pre-process** (upload STIL → **Run live simulation** → wait until **Done**).
- Step 2.** Open the **Post-process** tab.
- Step 3.** Add the ATE log from the chip you just tested (**Add log file** or **Add log folder**).
- Step 4.** Click **Analyze & recommend next chip**.
- Step 5.** Review the summary cards and the **Recommended patterns for next chip** table.
- Step 6.** Optionally click **Copy ordered list** and use that order on the next chip.

Analysis runs **locally in your browser**. Log files stay on your PC.

## 1. Full Post-process page

### test time optimization — Verilumen simulation agent

Pre-process: STIL → pattern prediction. Post-process: ATE logs → fail analysis.

Pre: STIL → predict · Post: logs → fail analysis

**Pre-process**  
STIL → pattern prediction

**Post-process**  
ATE logs → fail analysis

**POST-PROCESS**

**FAIL LOGS**  
After ATE: one log file = one die/chip. Analysis stays on this PC.

Add log folder

Add log files

chip\_N\_die42.log  
1 folder(s) · 1 file(s)

**Analyze & recommend next chip**

Clear logs

Next-chip: 57 to run (57 failed in logs) · 943 log pass-only skipped

Log folders (dies)  
1

high recommended patterns from pre-process  
29 across 29/600 pats

low risk patterns from pre-process  
28 across 28/400 pats

Files parsed  
1

**Recommended patterns for next chip**  
From your logs only: 1 log = 1 die/chip. Failed patterns are included; pass-only are skipped.

Recommend  
57 / 1000

Failed in logs  
57 must-run

Log pass → skip  
943

Failed patterns  
57

Copy ordered list

| Order | Pattern                    | Tier     | Fail rate | Fails | Why                                      |
|-------|----------------------------|----------|-----------|-------|------------------------------------------|
| 1     | P485 <span>was skip</span> | log fail | 100.0%    | 1     | failed on this chip · 1 channel STATUS:F |
| 2     | P70 <span>was skip</span>  | log fail | 100.0%    | 1     | failed on this chip · 1 channel STATUS:F |
| 3     | P7                         | log fail | 100.0%    | 1     | failed on this chip · 1 channel STATUS:F |

The page has a **header** (title + Pre/Post switch), a **left panel** (add logs and run analysis), and a **main area** (status, summary metrics, and the next-chip recommend table).

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## 2. Switching to Post-process

# test time optimization — Verilumen simulation agent

Pre-process: STIL → pattern prediction. Post-process: ATE logs → fail analysis.

Pre: STIL → predict · Post: logs → fail analysis

### Pre-process

STIL → pattern prediction

### Post-process

ATE logs → fail analysis

Under the title: **Pre-process** = STIL → pattern prediction (before ATE); **Post-process** = ATE logs → fail analysis (after ATE). You need a finished pre-process run so the tool knows the full pattern set (keep + skip, usually 1000 patterns).

## 3. Left panel — Fail logs

POST-PROCESS

FAIL LOGS

After ATE: one log file = one die/chip. Analysis stays on this PC. Add the log from chip N to recommend patterns for chip N+1.

Add log folder

Add log files

chip\_N\_die42.log  
1 folder(s) · 1 file(s)

Analyze & recommend next chip

Clear logs

Post-process ready — 1000 patterns from pre-process (keep 600 · skip 400). Add logs and Analyze.

Next-chip pattern recommendation

Pre-process predicts from STIL before ATE. Post-process uses real fail logs from tested chips, then recommends a reduced ordered pattern list for the next chip.

1. Complete Pre-process (STIL → Run live simulation)

2. Add log folders (chips/dies) from your PC

3. Click Analyze & recommend next chip

Use **Add log folder** or **Add log files**. For the usual flow (1 chip → 1 log), one log file is enough. **Analyze & recommend next chip** parses pass/fail and channel STATUS:F, matches patterns to the last pre-process keep/skip lists, and builds the ordered next-chip list. **Clear logs** removes selected files and clears the current analysis. Until you analyze, the main pane shows a short checklist.

Status may show keep/skip counts from pre-process (e.g. keep 600 · skip 400). That is context only — analysis still uses **all** patterns, not only the keep list.

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#### 4. Status line and summary cards

Next-chip: 57 to run (57 failed in logs) · 943 log pass-only skipped

|                         |                                                                     |                                                             |                   |
|-------------------------|---------------------------------------------------------------------|-------------------------------------------------------------|-------------------|
| Log folders (dies)<br>1 | high recommended patterns from pre-process<br>29 across 29/600 pats | low risk patterns from pre-process<br>28 across 28/400 pats | Files parsed<br>1 |
|-------------------------|---------------------------------------------------------------------|-------------------------------------------------------------|-------------------|

Log labels are 1-based (P1...P1000); tables use Verilumen agent ids (0...999). Fail = any channel STATUS:F on a die.

The status line reports how many patterns to run next, how many failed in the log, and how many pass-only patterns were skipped. Summary cards: **Log folders (dies)** (1 file → usually 1); **high recommended patterns from pre-process** (fails among kept patterns); **low risk patterns from pre-process** (fails among skipped patterns); **Files parsed**. These two fail cards compare silicon results to the pre-process split; they do not limit the next-chip list to keep-only. A skip pattern that failed can still be recommended (shown later with a **was skip** badge).

Log labels are often 1-based (P1...P1000); the dashboard uses ids 0...999 (log Pn → id n-1). Fail = any channel STATUS:F.

#### 5. Recommended patterns — header and metrics

**Recommended patterns for next chip**  
From your logs only: **1 log = 1 die/chip**. Every pattern that failed on that chip is included, ordered by severity (channels → fail rate → fails). Patterns that only passed are skipped.

Copy ordered list

|                        |                               |                        |                       |
|------------------------|-------------------------------|------------------------|-----------------------|
| Recommend<br>57 / 1000 | Failed in logs<br>57 must-run | Log pass → skip<br>943 | Failed patterns<br>57 |
|------------------------|-------------------------------|------------------------|-----------------------|

This is the main deliverable. Help text: **1 log = 1 die/chip**; every pattern that failed is included; order by severity (channels → fail rate → fails); pass-only patterns are skipped. **Copy ordered list** copies pattern ids in run order. Metrics: **Recommend** (to run / total), **Failed in logs**, **Log pass → skip**, and **Failed patterns**.

6. Recommend table (ordered list)

| Recommended patterns for next chip |                            |                       |           |       |                                                                                                                              |
|------------------------------------|----------------------------|-----------------------|-----------|-------|------------------------------------------------------------------------------------------------------------------------------|
| Order                              | Pattern                    | Tier                  | Fail rate | Fails | Why                                                                                                                          |
| 1                                  | P485 <span>was skip</span> | <span>log fail</span> | 100.0%    | 1     | failed on this chip (1 log = 1 die) · 1 fail(s), 100.0% fail rate · pre-process skip, failed on silicon · 1 channel STATUS:F |
| 2                                  | P70 <span>was skip</span>  | <span>log fail</span> | 100.0%    | 1     | failed on this chip (1 log = 1 die) · 1 fail(s), 100.0% fail rate · pre-process skip, failed on silicon · 1 channel STATUS:F |
| 3                                  | P296 <span>was skip</span> | <span>log fail</span> | 100.0%    | 1     | failed on this chip (1 log = 1 die) · 1 fail(s), 100.0% fail rate · pre-process skip, failed on silicon · 1 channel STATUS:F |
| 4                                  | P7                         | <span>log fail</span> | 100.0%    | 1     | failed on this chip (1 log = 1 die) · 1 fail(s), 100.0% fail rate · 1 channel STATUS:F                                       |
| 5                                  | P17                        | <span>log fail</span> | 100.0%    | 1     | failed on this chip (1 log = 1 die) · 1 fail(s), 100.0% fail rate · 1 channel STATUS:F                                       |
| 6                                  | P43 <span>was skip</span>  | <span>log fail</span> | 100.0%    | 1     | failed on this chip (1 log = 1 die) · 1 fail(s), 100.0% fail rate · pre-process skip, failed on silicon · 1 channel STATUS:F |

Each row is one pattern for the next chip, in suggested run order. Columns: **Order**, **Pattern** (badge **was skip** = pre-process skip list but failed on silicon), **Tier** (**log fail** = failed in the ATE log), **Fail rate** = fails÷(fails+passes), **Fails**, and **Why**.

With one die, order among failers is: more failing channels → higher fail rate → more fails → tie-breaks. If many rows look identical (all 100% fail rate, 1 fail), they are effectively a tie. Fail rate 100% on one die means fails/(fails+passes) = 1/(1+0) — failed on this chip, not “fails forever on every chip.”

7. Logic summary

- Chip N ATE log → parse each pattern (pass / fail / channel STATUS:F).
- Failed ≥ 1 time → INCLUDE (order by severity).
- Passed only → SKIP for next chip.
- Ordered list → run on Chip N+1.

8. Quick checklist

1. Pre-process: STIL → Run live simulation → Done
2. Open **Post-process**
3. Add log files (1 log = 1 die)
4. Analyze & recommend next chip
5. Read summary cards (high recommended / low risk fails from pre-process)
6. Use the recommend table order; **Copy ordered list** for the next chip

Source: docs/POST\_PROCESS\_GUIDE.md · Screenshots: docs/images/post-\*.png